

# Amutheezan Sivagnanam

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## SUMMARY

**PhD Candidate** and **Research Assistant** specializing in **Machine Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, **ML training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python, TensorFlow, PyTorch**, **AWS**, and agile CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

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## EDUCATION

### Pennsylvania State University

Ph.D., Computer Science — *Aug 2025*

### University of Houston

M.S., Computer Science — *Aug 2022*

### University of Moratuwa

B.S., (Hons) Engineering (Computer Science and Engineering) — *Jan 2018*

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## RESEARCH EXPERIENCE

### Pennsylvania State University — Graduate Research Assistant

#### *Applied Artificial Intelligence Lab*

*Aug. 2022 - May 2025*

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).
- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problems to address these challenges.
- Applied artificial intelligence-based solutions to tackle real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences and presented results at DOE and NSF meetings.

- Developed a deep reinforcement learning (DRL) method for proactive responder repositioning in emergency management, achieving response times 1000× faster while reducing operational delays.
- Applied DRL to solve online vehicle routing with advance booking, enabling real-time confirmations within seconds.

**The University of Houston** — Graduate Research Assistant

***Resilient Networks and Systems Lab***

*Sep. 2019 - Aug. 2022*

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).
- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problem statements to effectively address these challenges.
- Applied AI-based solution approaches to real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences and presented results at DOE and NSF meetings.
- Introduced heuristics that reduced annual energy costs by \$140K for public transit agencies operating mixed fleets of buses.
- Implemented a deep reinforcement learning approach that reduced operational costs by 20% by enabling online booking for traditionally offline Vehicle Routing Problems (VRPs).
- Collected and analyzed real-world data using Python to identify trends—such as changes in paratransit operations before and after COVID-19 and the impact of vulnerability reward programs.

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**SOFTWARE ENGINEERING EXPERIENCE**

**LSEG Technology** — Software Engineer

***Post Trade***

*Jan 2018 – Jul 2019*

- Writing application software in the Object-Oriented manner.
- Practicing agile development practices.
- Working experience with languages such as Java, Python, and C++
- Introduce Unit Testing for Libraries in Post Trade C++ Code
- Worked on Making changes in DB and Tested with BDD based Testing.

- Worked on Duplicated Error Codes Identification and Replacement with valid New Error Codes.
- Developed DSO for Front-End End to End Testing.
- Analyzed and Changed Back-End Regression Script to work with both Old and New Testing Framework
- Developed a New Report Generation for Testing Framework, which used to be emailed at the end of regression.
- Completed the Code Integration works related to Back-End Regression
- Updated automatic updates to auto-generated codes based on Database changes using Integration Plans.
- Participated in Code Integration and Code Deployment Plans.
- Participated in professional training programs conducted by Millennium IT Software and Post Trade Team.
- Worked on Front-End Development for both Product and Solution which consists of Enhancement, Bug Fixing, Merging and Introducing new features.

## **WSO2 Lanka PVT Ltd — Software Engineering Intern**

### ***Data Analytics Team***

*Jul 2016 – Dec 2016*

- Developed an alert generation system analyzing descriptive HL7/FHIR data to monitor disease outbreaks, triggering timely email and SMS notifications.
- Engineered a mechanism to evaluate hospital functionality by assessing admission and discharge messages, including bed and oxygen cylinder availability.

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## **SKILLS**

**Programming Languages:** Python, C++, Java, C, JavaScript, PHP, MATLAB

**Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Transformers, Deep Learning, Reinforcement Learning, DQN, DDPG, TRPO, PPO, GRPO, RLHF, Bayesian Optimization, Clustering, Statistical Analysis, Data Mining, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib

**Databases:** SQL, MySQL, MongoDB, SQLite

**Tools & Platforms:** Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash

**Methodologies & Practices:** Object-Oriented Development, Agile Practices, Behaviour Driven Testing

**Optimization & Operations:** Linear Programming, Integer Programming, CPLEX, Gurobi, Mosek, OR-Tools

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## PUBLICATIONS

### CONFERENCE PROCEEDINGS

- [C7] **Sivagnanam, A.**, Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the International Conference on Machine Learning (ICML-24)*.
- [C6] R. Sen, **A. Sivagnanam**, A. Laszka, A. Mukhopadhyay and A. Dubey, "Grid-Aware Charging and Operational Optimization for Mixed-Fleet Public Transit," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 4172-4179,
- [C5] Pavia, S., Rogers, D., **Sivagnanam, A.**, Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 8767–8770).
- [C4] Atefi, S., **Sivagnanam, A.**, Ayman, A., Grossklags, J., & Laszka, A. (2023, May). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference* (pp. 2209–2219).
- [C3] **Sivagnanam, A.**, Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).
- [C2] **Sivagnanam, A.**, Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938)*.
- [C1] Ayman, A., Wilbur, M., **Sivagnanam, A.**, Pugliese, P., Dubey, A., & Laszka, A. (2020, September). *Data-driven prediction of route-level energy use for mixed-vehicle transit fleets*. In *Proceedings of the 6th IEEE International Conference on Smart Computing (SMARTCOMP)* (pp. 1–8). IEEE.

### JOURNALS

- [J2] Wilbur, M., Ayman, A., **Sivagnanam, A.**, Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on Public Transit Accessibility and Ridership. *Transportation Research Record*, 2677(4), 531-546.
- [J1] Ayman, A., **Sivagnanam, A.**, Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2022). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology*, 22(1), Article 7, 1–29

## WORKSHOPS

[W1] **Sivagnanam, A.**, Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

## BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.**, Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.)**, *Artificial Intelligence and Society* (book chapter). ACM Press.

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## SERVICES

### Program Committee Member

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|-----------------------------------------------------------|------|
| International Joint Conference on Artificial Intelligence | 2025 |
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### Reviewer

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| International Joint Conference on Artificial Intelligence | 2024 |
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### Reviewer

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| AI4Research Workshop @IJCAI2024 | 2024 |
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### Auxiliary Reviewer

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| 22nd International Conference on Autonomous Agents and Multiagent Systems | 2023 |
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## AWARDS AND HONORS

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|----------------------------------------------------|------|
| <b>Information Security Quiz 2015 - Runners Up</b> | 2015 |
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Issued by Sri Lanka Computer Emergency Readiness Team(SLCERT)  
and Information and Communication Technology Agency of Sri Lanka(ICTA)

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| <b>Mahapola Scholarship</b> | 2014 - 2017 |
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